www.tntextbooks.in

where k is a dimensionless constant of proportionality. Rewriting above equation in terms of dimensions and taking k = 1, we have

$$[MLT^{-2}] = [M]^a[LT^{-1}]^b[L]^c$$
$$= [M^a L^b T^{-b} L^c]$$
$$[MLT^{-2}] = [M^a][L^{b+c}][T^{-b}]$$

Comparing the powers of M, L and T on both sides

$$a = 1; \; b + c = 1 \quad -b = -2$$
$$2 + c = 1 \quad b = 2$$
$$a = 1 \; b = 2 \text{ and } c = -1$$

From the above equation we get

$$F = m^a v^b r^c$$
$$F = m^1 v^2 r^{-1}$$
$$\text{or } F = \frac{mv^2}{r}$$

Limitations of Dimensional analysis

1. This method gives no information about the dimensionless constants in the formula like 1, 2, ……..π,e, etc.
2. This method cannot decide whether the given quantity is a vector or a scalar.
3. This method is not suitable to derive relations involving trigonometric, exponential and logarithmic functions.
4. It cannot be applied to an equation involving more than three physical quantities.
5. It can only check on whether a physical relation is dimensionally correct but not the correctness of the relation. For example using dimensional analysis, $s = ut + \frac{1}{3} at^2$ is dimensionally correct whereas the correct relation is $s = ut + \frac{1}{2} at^2$.